

Comparing the synthetic images generated on the various models:

The Vanilla Autoencoder struggled when detecting outlines on both the numerical and fashion dataset. The model is very simple with only having 1 hidden layer so it produces images that look like blocks or chunks of objects and often result in blurred output images. It struggles with understanding shapes and when trying to ascertain specific elements within the fashion dataset like patterns, colors and logos the model fails to do so accurately. When trained on the combined dataset the Vanilla Autoencoder produces images that are very hard to recognize from the original image.

The Variational AutoEncoder did a good job at reconstructing the numerical MNIST dataset but when it came to the fashion dataset the model struggled to pick up on the specific small details within the clothing items. Elements such as clothing patterns, logos, and color nuances were noticeably not existent. Despite this, the model still demonstrated a strong ability to capture the overall shape and outline of the clothing items, effectively identifying their general form. This suggests that while the Variational Autoencoder is well-suited for simpler, low-detail datasets, it lacks the ability to encode and reconstruct the higher-level features required for more intricate visual data. When trained on the combined dataset we noticed that the model did well at recreating the edges and outlines of the images but the detail quality in the fashion images declined even further.

The Generative Adversarial Networks (GAN) struggled with accurately recreating the outlines of images, yet they consistently produced the highest-quality results overall. The images generated typically had, feature correct color schemes, detailed logos, and accurate patterns in fashion items. When examining the numerical dataset, it became apparent that the model failed to capture the image outlines with precision. Even in the fashion dataset, issues such as missing sleeves or incomplete clothing items were common. We determined that the GAN was able to produce the best quality images due to the adversarial training architecture. The generator is forced to increasingly produce more realistic outputs in order to fool the discriminator. As a result, the GAN learns to replicate textures, colors, and intricate details such as logos and patterns more effectively than other models. Training on the combined dataset led to a noticeable decline in performance. Not only did the outlines of the images become less distinct, but the overall image quality also deteriorated. The outline detection is especially noticeable when comparing the results from the combined model's numerical image generation with the

single model's numerical generation. Similarly, in the fashion dataset, the logos and patterns produced by the combined model appeared washed out and less defined.

We faced multiple challenges when training the GAN compared to the auto encoders. Some of these challenges include:

- Rapid switches between high generator loss and high discriminator loss, lead to unstable training.
- Autoencoders typically maintained steady convergence in reproducing quality but GAN's are highly variable in loss metrics.

Steps taken to increase accuracy:

- Adjustment of learning rates, mostly slowing down the discriminator and increasing generator speed to balance training.
- Changed to using fixed noise to reduce variability.
- Added multiple discriminator updates per generator update to stabilize D_Loss and increased D_accuracy as a result.

The latent space visualizations is helpful in understanding VAES because it provides insight into the model's ability to classify or categorize the reconstructed images based on their latent representations (z-scores). By mapping these z-scores, we can observe how the model organizes different the classes of the data which helps determine how well separated the categories are in a latent space and where overlaps occur. The overlaps shown in the the laten space visualizations are areas where the model may struggle to differentiate between similar classes, which can lead to misclassifications or blurry reconstructions of images.